UNIVERSITY COLLEGE DUBLIN

SCHOOL OF ELECTRICAL & ELECTRONIC ENG.

EEEN40010 CONTROL THEORY

Laboratory CT_1

DOMINANT POLE PLACEMENT VIA ROOT LOCUS

1. Objective:

To practice the control design method of dominant pole placement via root locus.

Before starting this laboratory generate some personalised digits. Let a, b, c be the last three digits of your student number. For example, if your student number is 19306227 then a = 2, b = 2 and c = 7. If any of these digits is 0 then add 1. For example, if your student number is 19306207 then a = 2, b = 1 and c = 7.

The problems which you are to solve are shown in blue. You must produce a report for this laboratory. This report will be graded and that grade will carry a weight of 33% when calculating your overall module grade. This is a slight reduction on previous years in light of student feedback. The number of grade steps which can be earned for each of the problems is given in the problem statement and a coarse grading scheme follows each problem statement.

In designing using root loci you should generally sketch a few root loci to see the expected shape for choices of various parameters (i.e. various introduced poles and zeros). If sketched by hand, root loci should be subsequently scanned/photographed and included as an image in the report. This is the **single** exception to the general rule regarding acceptability of handwritten material. It is generally not acceptable, but here the requirement is for you to reveal your understanding of the root locus drawing rules by applying them and assessors cannot judge that understanding if you have the MATLAB/Python command/method **rlocus** draw those root loci for you. You may alternatively, if you wish use sketching software to produce these images of "sketched" root loci for your report. To reiterate, what is sought here is evidence that you understand and can apply the root locus drawing rules, because this skill is useful in controller design using the root locus method.

2. Controller Design:

Problem 1: (9.25 grade steps)

The longitudinal dynamics of an aircraft in steady, level flight are approximately described by the following equations of motion:

$$\dot{\tilde{\alpha}} = a_{11}\tilde{\alpha} + a_{12}\tilde{q} + b_1\tilde{\delta}_e + b_1 d$$
$$\dot{\tilde{q}} = a_{21}\tilde{\alpha} + a_{22}\tilde{q} + a_{23}\dot{\tilde{\alpha}} + b_2\tilde{\delta}_e + b_2 d$$
$$\dot{\tilde{\theta}} = \tilde{q}$$

(2)

where $\tilde{\alpha}$ is the offset of the angle of attack (i.e. the difference between this and the desired value or operating value), $\tilde{\theta}$ is offset of the pitch angle (i.e. again the difference between this and the operating value), $\tilde{\delta}_e$ is the elevator setting and d is a disturbance. Take the following highly non-

realistic values for parameters: $a_{11} = -1.a$, $a_{12} = -b$, $a_{21} = 0.ab$, $a_{22} = -1.ac$, $a_{23} = 0.00a$, $b_1 = 0.cba$, $b_2 = 1.abc$. For example, $0.ca$ is the decimal number for which c is the first digit after decimal point and a is the second digit where **a**, **b**, **c** are your personal digits. Show that the plant has one real negative zero, one real pole and one complex conjugate pair of poles. Show also that the complex conjugate pair of poles lie to the left of the plant zero in the complex plane. Determine the time constant of the plant zero. By employing dominant pole placement via the root locus and presenting your design decisions and methodology in detail, design a controller which such that:

(i) the closed-loop system is stable
(ii) there is zero steady state error to a step input
(iii) the 2% settling time does not exceed twice the time constant of the plant zero but is not less than one third of this time constant. These two specifications should not be in conflict, but in the very unlikely event that they are the first specification should take precedence.
(iv) the PO% does not exceed 80% of the percentage overshoot of the canonical second order system whose poles equal the complex conjugate pair of plant poles or does not exceed 30%, whichever is larger.

Your controller should have a relative degree not less than 0, as otherwise it is not realisable. Your design should employ a controller with coefficients which are small where possible, as unnecessarily large coefficients yield poorer performance in general. You should give a full report on your "topology discovery". Upon completion of your design simulate the closed-loop system to determine whether design specifications have been achieved, at least in the ideal simulation environment.

Although you have not designed to achieve good disturbance rejection, estimate the disturbance rejection nonetheless by considering the response of the system when the reference input is set to 0 and when the disturbance d is a small impulse. Characterise this disturbance response by finding its maximum modulus and by estimating its duration.

A control design **should** have robust performance, meaning that it should continue to approximately work in spite of inaccuracies such as incorrect modelling of the plant, provided those inaccuracies are not too large.

Investigate the robustness of your control design by considering the performance of the closed-loop system in the event where your modelling of the plant has been incorrect and the true values of parameters a_{21} and a_{11} differ from their nominal values by as much 10%.

A control design **must** have robust stability, meaning that it must continue to be closed-loop stable in spite of inaccuracies such as incorrect modelling of the plant, provided those inaccuracies are not too large. By offering appropriate evidence, indicate whether or not the closed-loop system which you have designed should be considered to be robustly stable.

Grading Scheme: Conversion from specifications to desired dominant pole(s) and requirement or otherwise of I-term (1.5 grade steps), controller design using root locus (4 grade steps), numerical closed-loop performance and comparison with specifications (0.75 grade steps), investigate disturbance rejection (1 grade step), investigate robustness of performance to poorly modelled plant (1 grade step), investigate robustness of stability (1 grade step).

Problem 2: **(11.75 grade steps)**

A SISO plant with input f and output x is described by the following ODE:

$$\frac{d^2x}{dt^2} - (\alpha_1 - \alpha_2 x^2)\frac{dx}{dt} + \beta_1 x + \beta_2 x^3 + \beta_3 x^5 = f(t) \quad (2)$$

Let $\alpha_1 = 1 + \frac{(a+10)}{25}$, $\alpha_2 = \frac{\alpha_1}{3}$, $\beta_1 = 1 + \frac{b}{10}$, $\beta_2 = \frac{c}{25}$ and $\beta_3 = \frac{a+b}{100}$ where **a**, **b**, **c** are your personal digits. Setting the forcing term *f*(*t*) to 0, we obtain a special case of the Liénard equation. Confirm numerically that the system settles to a stable oscillation, i.e. a limit cycle (provided initial conditions are not $x = 0,\ \dot{x} = 0$). Determine the period of this oscillation. The nominal output *x* and the nominal input *f* are both taken to be 0. The desired operating point is: $x = 0,\ \dot{x} = 0$. Obtain a linearised model of the system in the locality of this operating point, presenting full details of your analysis in the report. Determine the transfer function of this local linear, constant coefficient model. Confirm that this linearised system is unstable, providing evidence for this conclusion.

By employing dominant pole placement via root locus design a practical PID controller (i.e. a 4-term controller where the D-term, if included, is not an ideal differentiator but rather incorporates the filter $1/(T_D s+1)$ considered in lectures) such that:

(i) the closed-loop system is stable
(ii) there is zero steady state error to a step input
(iii) the 2% settling time does not exceed 85% of the period of oscillation of the unforced nonlinear plant.
(iv) the PO% does not exceed 80%

In the report present clear reasons for each of your design decisions, your "topology discovery" and present the full details of the design. Determine the step response of the resulting linear closed-loop system and determine whether design specifications have been achieved, presenting data to support this determination.

Presenting your reasoning determine whether the plant should be deemed failsafe relative to the likely failure mechanisms of the controller.

Determine and present the equations of motion of the actual closed-loop system incorporating the PID controller and the nonlinear global model of the plant given in Eq. (2). Hint: unless a P-term only is present in setting up these equations of motion you will have to model both the plant and the controller, and subsequently the whole closed-loop system using the state-space models which were very briefly discussed through a single example in CT_0 and which will be discussed in far more detail later in lectures. By numerically solving the equations of motion of the nonlinear closed-loop system confirm that the linearised closed-loop system and the actual closed-loop system have very similar step responses provided the size of the step input is very small. Present and discuss the code used to numerically solve. Also, determine how the steady-state error, the settling time and the overshoot are affected by increasing the size of the step input.

Grading Scheme: Confirm oscillation and estimate period (0.5 grade step), obtain linearised local model, determine transfer function, confirm unstable (1.5 grade steps), conversion from specifications to desired dominant pole(s) and requirement or otherwise of I-term (1.5 grade steps), controller design using root locus (4 grade steps), numerical closed-loop performance and comparison with specifications (1 grade step), explain whether or not the system is failsafe (0.5 grade step), obtain global model of nonlinear closed-loop system (1 grade step), confirm agreement of local and global closed-loop models for small step input, including code and discussion of detailed analysis used (1 grade step), determine the effect of the size of the step input on performance characteristics (0.75 grade steps).